

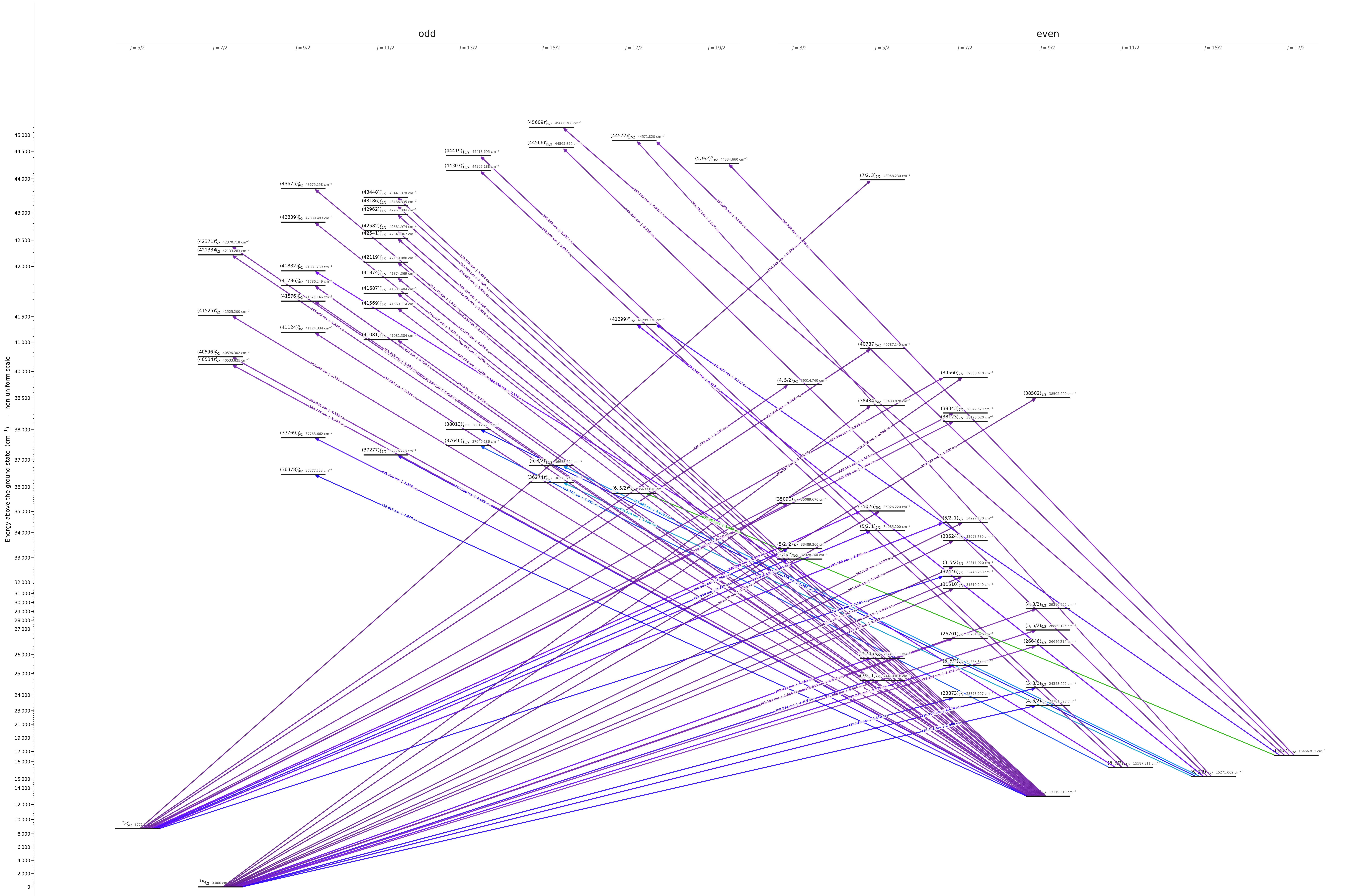
Thulium

Tm (Tm I). Strongest classified transitions below 2 μm (NIST): 70 levels, 70 lines.

Tm⁺ ionisation limit 49880.57 cm⁻¹ = 6.18440 eV (200.479 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle J || e r || J' \rangle$ in ea₀
Matrix elements are measured or NIST values: * = high-accuracy theory, = = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: NIST energy and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle J e r J' \rangle$ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
(6, 3/2) _{1/2} - (42962) _{1/2}	335.0951	334.9989	894.6489	3.835	6.600e+07	-	-	-
(6, 3/2) _{1/2} - (44566) _{1/2}	343.3570	343.2591	878.2374	6.339	1.200e+08	-	-	-
(6, 5/2) _{1/2} - (45609) _{1/2}	343.0312	342.9329	873.9510	6.002	1.130e+08	-	-	-
(6, 3/2) _{1/2} - (41874) _{1/2}	347.7685	347.6690	862.0460	4.085	6.700e+07	-	-	-
(6, 3/2) _{1/2} - (41786) _{1/2}	348.8375	348.7377	859.4042	5.790	1.600e+08	-	-	-
(6, 5/2) _{1/2} - (44572) _{1/2}	355.6832	355.5816	842.8637	5.097	6.500e+07	-	-	-
(6, 5/2) _{1/2} - (5, 9/2) _{1/2}	358.7090	358.6067	835.7538	5.688	7.100e+07	-	-	-
(6, 3/2) _{1/2} - (40596) _{1/2}	363.9448	363.8411	823.7305	4.555	1.090e+08	-	-	-
²F₇/₂ - (5, 5/2) _{1/2}	371.8976	371.7918	806.1157	6.046	1.440e+08	-	-	-
²F₇/₂ - (26701) _{1/2}	374.5132	374.4068	800.4856	4.531	9.900e+07	-	-	-
²F₇/₂ - (35090) _{1/2}	379.9619	379.8541	789.0066	3.650	1.230e+08	-	-	-
²F₇/₂ - (25745) _{1/2}	388.4232	388.3131	771.8192	4.289	1.060e+08	-	-	-
²F₇/₂ - (5/2, 1) _{1/2}	391.7585	391.6476	765.2480	6.856	1.980e+08	-	-	-
²F₇/₂ - (5/2, 1) _{1/2}	395.0390	394.9272	758.8933	5.001	1.370e+08	-	-	-
²F₇/₂ - (7/2, 1) _{1/2}	409.5236	409.4181	732.0238	4.465	9.800e+07	-	-	-
²F₇/₂ - (3, 5/2) _{1/2}	413.9498	413.8331	724.2241	3.326	7.900e+07	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ).